

Microtubule dynamics: an interplay of biochemistry and mechanics

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Abstract | Microtubules are dynamic polymers of $\alpha\beta$ -tubulin that are essential for intracellular organization, organelle trafficking and chromosome segregation. Microtubule growth and shrinkage occur via addition and loss of $\alpha\beta$ -tubulin subunits, which are biochemical processes. Dynamic microtubules can also engage in mechanical processes, such as exerting forces by pushing or pulling against a load. Recent advances at the intersection of biochemistry and mechanics have revealed the existence of multiple conformations of $\alpha\beta$ -tubulin subunits and their central role in dictating the mechanisms of microtubule dynamics and force generation. It has become apparent that microtubule-associated proteins (MAPs) selectively target specific tubulin conformations to regulate microtubule dynamics, and mechanical forces can also influence microtubule dynamics by altering the balance of tubulin conformations. Importantly, the conformational states of tubulin dimers are likely to be coupled throughout the lattice: the conformation of one dimer can influence the conformation of its nearest neighbours, and this effect can propagate over longer distances. This coupling provides a long-range mechanism by which MAPs and forces can modulate microtubule growth and shrinkage. These findings provide evidence that the interplay between biochemistry and mechanics is essential for the cellular functions of microtubules.

End-binding (EB) proteins

A family of evolutionarily conserved proteins that autonomously track the growing microtubule end. EB proteins regulate microtubule dynamics and are also responsible for recruiting the majority of other plus-end tracking proteins to the growing microtubule end.

Microtubules are long, stiff polymers of $\alpha\beta$ -tubulin that are structurally and functionally important components of the eukaryotic cell cytoskeleton, forming the mitotic spindle and the axonemes of cilia and flagella and serving as tracks for intracellular trafficking¹. Microtubules are highly dynamic. They grow by the addition of GTP-tubulin dimers to the microtubule end, where a stabilizing cap forms. Below the cap, tubulin dimers are in the GDP-bound state, owing to GTPase activity of tubulin². This cap recruits end-binding (EB) proteins. A host of other microtubule-associated proteins (MAPs) associate with the microtubule end, including microtubule polymerases, microtubule depolymerases and regulatory kinesins, which cooperate and compete to determine whether the microtubule grows or shrinks^{3–5}. When the stabilizing cap is lost, the microtubule shrinks rapidly in an event known as a catastrophe. Occasionally, a rapidly shrinking microtubule is rescued, switching from shrinkage back to growth. This switching behaviour, known as dynamic instability⁶, can be harnessed by cells to exert forces, for example, during chromosome segregation in mitosis⁷.

Biochemical and mechanical phenomena are intimately intertwined in microtubules: rapid turnover and dynamic instability coexist with the rigidity of a stiff polymer⁸. This interplay between biochemistry and mechanics allows microtubules to be viewed from multiple perspectives. A biochemical perspective views microtubules as a

collection of discrete subunits⁹ that form a lattice with two main modes of interaction: longitudinal interactions between tubulins make up the protofilaments, which are held together by lateral interactions (FIG. 1a,b). Recent developments in single-molecule microscopy¹⁰ enable tracking of microtubule growth with a resolution of ~40 nm (REF.¹¹); this has enabled characterization of the behaviour of the stabilizing cap immediately before a catastrophe¹² and of the stochastic fluctuations in growth rate that occur during elongation^{13,14}. Recent measurements such as these provide new insights into the biochemistry that underlies microtubule dynamics. A mechanical perspective, on the other hand, views microtubules as a continuous, stiff material⁹. Indeed, seminal measurements showed that microtubule stiffness is comparable to that of plexiglass¹⁵.

The distinction between biochemical and mechanical perspectives is artificial⁹, of course, because mechanical phenomena operate at the level of discrete subunits (FIG. 1). The biochemical cycle of tubulin involves GTP-tubulin binding to the microtubule end, hydrolysing the GTP and releasing phosphate while in the lattice and then dissociating from the microtubule as GDP-tubulin. Notably, these biochemical transitions are accompanied by conformational changes. In this Review, we outline the biochemical and conformational cycles of tubulin and their roles in microtubule dynamics. We advance the argument that

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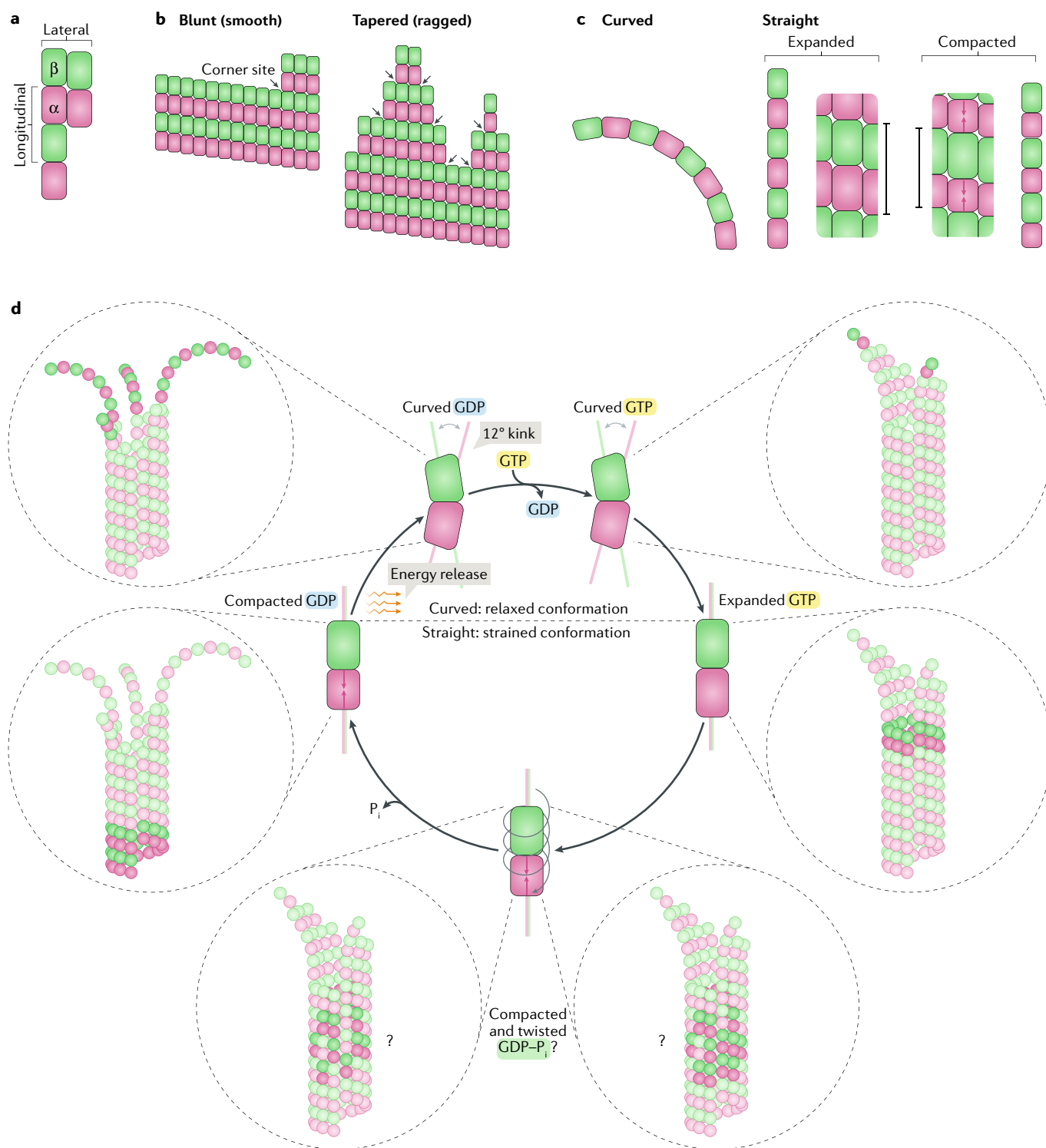


Fig. 1 | Tubulin biochemistry, conformation and mechanics. a | Cartoon illustrating the fundamental longitudinal (vertical) and lateral (horizontal) interactions between the $\alpha\beta$ -tubulin dimers that make up the microtubule lattice. **b** | Two-dimensional schematics of blunt (few corner sites, arrows) and tapered (more corner sites, arrows) microtubule ends. Tapering refers to the extension of some protofilaments beyond others; raggedness describes an uneven or rough distribution of protofilament lengths. **c** | Cartoons of longitudinal assemblies of $\alpha\beta$ -tubulin illustrating the three major conformations in structural detail. Tubulin adopts a curved conformation in single protofilaments and when unpolymerized (left). Tubulin adopts straight expanded or compacted conformations in the body of the microtubule (right).

The horizontal bars indicate that the compacted conformation is somewhat shorter than the expanded conformation. Compaction results from reordering of the α -tubulin subunit (arrows). **d** | The tubulin conformation cycle is also a mechanical cycle. Straight conformations of $\alpha\beta$ -tubulin are strained but stabilized by interactions with the microtubule lattice (not illustrated here). Release of that strain during depolymerization can be used to perform mechanical work. The thumbnail cartoons of growing or shrinking microtubules indicate the region of the microtubules where the different conformations occur. Currently it is unclear (indicated by '?') whether the compacted and twisted conformation reflects a specific nucleotide state (GDP- P_i) and exactly how these tubulin conformations are distributed in the lattice.

Microtubule polymerases

Regulatory factors that promote microtubule elongation, typically by making microtubules grow more quickly.

Microtubule depolymerases

Regulatory factors that promote microtubule depolymerization. Well-studied microtubule depolymerases include the mitotic centromere-associated kinesin (MCAK)–kinesin 13 and the Kip3–kinesin 8 families.

Kinesins

A large family of microtubule-associated proteins that use the energy from ATP hydrolysis to walk on microtubules and/or regulate their polymerization dynamics.

Plus end

The end of the microtubule that is crowned by β -tubulin subunits. This more dynamic end of the polymer typically grows from the centre of the cell towards the periphery.

Minus end

The end of the microtubule that is crowned by α -tubulin subunits. This less dynamic end of the polymer is typically capped or anchored in a microtubule-organizing structure such as the centrosome.

changes in tubulin conformation that accompany this biochemical cycle are mechanical in nature. This concept of a relationship between biochemical transitions and mechanics is found throughout cell and molecular biology because conformational changes associated with nucleotide hydrolysis are harnessed to do mechanical work by a wide range of oligomeric proteins (see REFS^{16,17} for some recent examples). We argue that microtubules represent a unique example of mechanochemical coupling because of the structural complexity of their growing and shrinking ends. First, we discuss the dynamics of the microtubule end and describe evidence that tubulin dimers in the growing end transition through multiple conformations in response to the biochemical cycle of GTP hydrolysis. We also describe new data that emphasize the importance of the tubulin conformational cycle for understanding how microtubule dynamics generate forces and how forces can affect microtubule dynamics. We then discuss data that make it increasingly apparent that the conformational cycle of tubulin dimers is targeted by regulatory factors that control microtubule dynamics, providing additional evidence for the intimate links between mechanical and biochemical processes in microtubule dynamics and regulation¹⁸. Finally, we propose that long-range coupling between tubulin subunits may be the key to understanding microtubule dynamics because it provides a framework that integrates mechanical and biochemical properties within the lattice.

Tubulin cycle at the microtubule end

The transition of a GTP-tubulin dimer in solution into a GDP-tubulin dimer integrated into the microtubule lattice is a complex process that involves multiple conformational changes. Below, we describe the structure where this transition occurs, the microtubule end, and the mechanical phenomena that underlie this process.

Our focus is on the more dynamic plus end of the microtubule; the unique properties and regulation of the minus end have been reviewed elsewhere⁴.

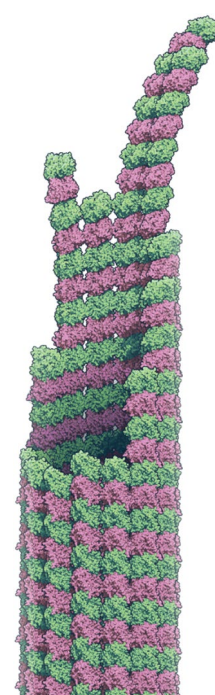
Structure of the microtubule end. For growing microtubules, ‘the end’ is often synonymous with the stabilizing cap, which describes the extended region near the growing end that is enriched in GTP-tubulin subunits. The stabilizing cap contains several hundred tubulin dimers¹⁹, but the exact distribution of nucleotide states within tubulin dimers integrated into the cap is currently not known. Using cryo-electron microscopy (cryo-EM), the growing microtubule end has been observed to adopt a range of structures, from short, blunt ends²⁰ to long, outwardly curved and flattened structures interpreted as ‘sheets’ of tubulin²¹. As a result, ‘the microtubule end’ remains poorly defined. Our lack of certainty about this structure can be observed in the diverse schematics of microtubule ends that appear in reviews, research articles and textbooks. The illustrations in these schematics range from large, elaborate sheets to simple, tapered structures that neglect the curvature of microtubule ends entirely. The most common schematics depict the growing microtubule end as a funnel. These funnels appear in undergraduate textbooks²² and on conference posters. While scientists are not necessarily artists, and we differ in our tolerance for flourish, it is fair to say that the field has no consensus about what a microtubule end actually looks like.

The major features of microtubule ends are nevertheless consistent across the literature: some protofilaments are longer than others; the distal protofilaments curve outward to some degree; and there is a ‘seam’ where lateral interactions are heterotypic (β -tubulin contacts α -tubulin) rather than homotypic (β - β or α - α) and thus perhaps weaker^{23,24}. BOX 1 describes our best-faith

Box 1 | Model for a growing microtubule end and how it was constructed

The structure of the microtubule end is currently elusive. Here, we show our best-faith effort to construct an atomic model of a growing microtubule end (see figure). We considered two features that distinguish growing microtubule ends from the rest of the lattice, for which atomic models exist¹²⁷. First, microtubule ends are tapered, with a taper length defined as the distance between the longest protofilament and the fully closed lattice. We chose a taper length of 124 nm, which is consistent with the average value measured directly by cryo-electron microscopy (cryo-EM)²⁵ and with estimates based on subpixel fitting of fluorescence images^{11,13}. Second, microtubule ends are variably curved owing to the fact that GTP-tubulin adopts a curved conformation. Our model was informed by a theory whereby microtubule ends are shaped by two opposing forces: the tendency to curve outward due to the intrinsic curvature of GTP-tubulin^{30,33–35} and the tendency to straighten due to lateral interactions between subunits that close up the lattice¹²⁵. The precise relationship between the curvature of the tip and its width is not understood. We assumed that a section of the microtubule end becomes fully straight²⁴ when it is at least six $\alpha\beta$ -tubulins wide²⁵, sections of the microtubule end that are two $\alpha\beta$ -tubulins wide are approximately half-straightened²⁶, and sections of the microtubule end that are three to five subunits wide become progressively straighter¹²⁵. Conformations with curvature in between half-straight²⁶ and straight²⁴ were approximated by interpolation (morphing). Finally, we added some ‘raggedness’ to the end, a feature that occurs frequently in computer simulations of microtubule dynamics^{27,28}.

Our final model used an average value for taper length. The stochastic nature of subunit addition and loss will cause the taper length and overall curvature of growing microtubule ends to fluctuate considerably around their average values. These and other uncertainties highlight the challenge of understanding how the biochemical and mechanical properties of individual $\alpha\beta$ -tubulin subunits dictate the structure and dynamics of the growing microtubule end.



Interpolation (morphing)

A computational procedure that creates a series of structural intermediates between two known conformations of a molecule. Conformations generated by morphing provide a useful way to visualize intermediate steps along a conformational change but do not always capture the actual pathway for the conformational change.

Strain energy

Describes internal energy stored by a molecule when that molecule is held in an otherwise unfavourable conformation.

Compressive stress

Strain energy resulting from forces that push against the growing microtubule end.

Tensile stress

Strain energy resulting from forces that pull on the growing or shrinking microtubule end.

Optical tweezers

Device in which radiation pressure from a laser beam focused by a microscope objective is used to trap a small bead or other object. Optical tweezers provide a sensitive way to measure both the position of the trapped object and the forces experienced by the trapped object.

XMAP215 family

The best-studied family of microtubule polymerases. Polymerases in this family, which includes vertebrate XMAP215 and yeast Stu2, are the major factors that promote fast microtubule growth in cells, and their activity is important for proper formation of the mitotic spindle.

Free energy

A thermodynamic quantity that describes the amount of energy that can do work in a given system. The change in free energy associated with a chemical reaction provides the amount of work that is released by that reaction or that must be inputted to drive that reaction.

effort to illustrate average values for taper length and protofilament curvature at the microtubule plus end that are consistent with findings from cryo-EM^{25,26} computational modelling^{27,28} and fluorescence microscopy^{11,13}. The data indicate that the microtubule end is tapered, ragged and variably curved, even if the specifics of these features remain poorly determined.

Tubulin dimers in the cap undergo a mechanochemical cycle. Individual dimers pass through at least three conformations during microtubule growth and shrinkage (FIG. 1c,d). These conformations, based on recent cryo-EM reconstructions^{24,29} and crystal structures³⁰, are described as curved, expanded and compacted (FIG. 1c). We argue that these conformations can be considered as mechanical states because they are part of a cycle that stores and releases strain energy^{31,32} (FIG. 1d). Unpolymerized GTP-tubulin dimers start out in the curved conformation^{33–35} (reviewed in REF.¹⁸), which is characterized by a $\sim 12^\circ$ kink at the intradimer interface. After binding to the microtubule end, the dimers straighten into an expanded conformation, a process that is thought to introduce strain energy into the microtubule lattice because the tubulin dimers are held in an otherwise unfavourable conformation. Following GTP hydrolysis and phosphate release, the dimers compact in the lattice, shortening in length by 3 Å through a movement of a subdomain of α -tubulin²⁹ (FIG. 1c,d). This understanding of compaction is based on structural studies of mammalian microtubules^{24,29}, but it is not yet clear whether microtubules from all species behave similarly. Indeed, recent structures of yeast microtubules did not show compaction of the GDP lattice^{36,37}. Regardless, after catastrophe, GDP-tubulin dimers relax back to the curved state, peeling outward into the ‘ram’s horns’ observed by cryo-EM²⁰. This relaxation releases the strain energy stored in the dimer, which can be harnessed to perform work³², and completes the mechanochemical cycle (FIG. 1d).

Mechanics in the tubulin cycle

Direct evidence that the conformational cycle of tubulin is tightly linked to mechanics can be found in studies of how microtubules generate and respond to force. A growing microtubule generates a pushing force when it encounters a barrier³⁸; this pushing force can be used to centre various structures, including microtubule asters³⁹, mitotic spindles⁴⁰ and nuclei⁴¹. Conversely, a shrinking microtubule generates a pulling force on objects that are ‘coupled’ to the microtubule end. The signature ‘coupler’ is the kinetochore³², a large molecular machine that attaches chromosomes to the growing and shrinking ends of a microtubule. Pulling forces in the spindle arise from the outward peeling of protofilaments, which either push directly on the coupler⁴² or bias its diffusion⁴³ on the microtubule lattice. Importantly, compressive stress and tensile stress generated by these pushing and pulling forces can affect the dynamic instability of microtubules themselves^{38,44}.

The response to tensile stresses generated by pulling forces provides a clear example of forces altering the mechanical state of tubulin. Tensile stresses occur when a shrinking microtubule pulls on a coupler or, conversely,

when the coupler pulls on a growing microtubule end. Both cases of tensile stress can be reconstituted in vitro and studied using optical tweezers. The couplers in these in vitro assays have included direct linkages⁴⁵ and purified kinetochore proteins^{43,45–47}, as well as intact kinetochore particles from budding yeast⁴⁴. Remarkably, applying tension to a growing microtubule via a bead with an intact kinetochore particle attached to it causes the microtubule to grow more quickly⁴⁴. Similarly, applying tension via a bead coated with a microtubule polymerase in the XMAP215 family also caused faster growth⁴⁸, although other coupling proteins, such as the Ndc80 complex, did not⁴⁶. Perhaps tension promotes microtubule growth by somehow enhancing the catalytic cycle of XMAP215 family polymerases? Kinetochores do contain a subunit with microtubule polymerase activity (for example, in budding yeast the polymerase is Stu2 (REF.⁴⁹), an XMAP215 homologue), but, surprisingly, kinetochore particles completely depleted of Stu2 retain the ability to stimulate microtubule growth in the presence of tension⁵⁰. So why does tension applied through intact kinetochore particles lead to faster microtubule growth? It could be that tension applied to microtubule ends alters their taper and curvature, promoting straighter ends with more stable lateral contacts. Pulling on a microtubule might also antagonize the compaction of expanded dimers in the stabilizing cap to counteract GTP hydrolysis (FIG. 1d). It remains unclear how such changes in tubulin conformation would lead to faster growth, but we attempt to shed some light on this matter in the final main section of the article. It is also elusive as to why intact kinetochore particles and XMAP215 have this positive effect on microtubule growth while other couplers do not.

The key to understanding the relationship between microtubule dynamics and forces will be continued measurements of the relationship between the two. As with the experiments above, optical trapping currently provides the best way to perform such measurements. In a recent study, the amount of work performed by a shrinking microtubule was measured using optical tweezers⁵¹, building upon a related assay pioneered by others⁴². In this assay, a bead bound to the side of a microtubule is maintained under tension in the direction of microtubule growth. When the depolymerizing microtubule end reached the bead, it ‘pushed’ the bead against the applied tension; the magnitude of the resulting deflection reflects the shape of the curling protofilaments. Interestingly, increasing the tension applied to the bead led to reduced deflections, indicating that tension was not allowing the outwardly peeling protofilaments to become fully curved. The work output per tubulin dimer was estimated at $\sim 4.7 k_B T$ (k_B is Boltzmann’s constant; $4.7 k_B T$ equates to 19 pN nm or 2.7 kcal mole⁻¹), which is approximately 25% of the free energy available from GTP hydrolysis. This efficiency of converting free energy into work is similar to that demonstrated for molecular motors such as kinesin and dynein^{52,53}. It is commonly assumed that what delivers energy for work is the release of the strain energy stored in compacted tubulin and that the release of this energy is the reason why microtubules shrink rapidly after catastrophe. However, microtubules harbouring

Box 2 | Recombinant $\alpha\beta$ -tubulin: a new tool to study microtubule dynamics

The current understanding of microtubule biochemistry has relied almost exclusively on $\alpha\beta$ -tubulin derived from mammalian (mostly bovine) brain preparations. While such brain preparations have large yields, the resulting $\alpha\beta$ -tubulin is a mixture of multiple α -tubulin and β -tubulin isotypes (see below) and varies in its degree of post-translational modifications¹²⁸.

Obtaining recombinant tubulin

The path to recombinant $\alpha\beta$ -tubulin began with its purification from genetically tractable model organisms, including *Saccharomyces cerevisiae* and *Tetrahymena thermophila*^{129,130}. Subsequent work showed that fusing an affinity tag to β -tubulin could improve purification without compromising function¹³¹ and that tagged tubulin still retains isotype-specific polymerization dynamics¹³². The binding sites for taxol¹³³ and kinesin^{134,135} were also probed using recombinant tubulin. The potential value of recombinant tubulin was clear.

These early recombinant expression methods had poor yields and were restricted to mutants that did not perturb polymerization dynamics too much. Inducible overexpression in yeast¹³⁶ improved yields and expanded the scope of purifiable mutants to include nonviable mutants. Methods for purifying recombinant human $\alpha\beta$ -tubulin from baculovirus-infected insect cells soon followed^{137–139}.

Illuminating mechanisms of microtubule-associated proteins and motors

A polymerization-blocked mutant of yeast tubulin¹³⁶ led to structures of complexes between $\alpha\beta$ -tubulin and the TOG1 or TOG2 domains from Stu2 (see main text)^{81,82}. Recombinant point mutants of tubulin created microtubules that separated the motility and depolymerase activities of the kinesin 8 family Kip3 (REF.⁷³). Motor proteins generally interact with the unstructured C termini of $\alpha\beta$ -tubulin, which previously could be removed only by proteolytic cleavage. Recombinant chimeric microtubules in which human C-terminal tails were fused to a yeast tubulin core demonstrated that specific tubulin tails or post-translational modifications affected different motor proteins in distinct ways¹⁴⁰.

Revealing properties of individual tubulin isotypes

Humans have seven α -tubulin genes and eight β -tubulin genes, some expressed only in specific cell types. Studies with recombinant, single-isotype mammalian $\alpha\beta$ -tubulin discovered different polymerization dynamics for the isotypes relative to bovine brain tubulin^{138,139,141}, echoing observations for different yeast isotypes¹³². Although $\alpha\beta$ -tubulin isotypes differ mainly in the sequence of their C-terminal tails, in at least one case the differences in dynamics could be attributed to sequence variation in the more conserved structured core of the protein¹⁴¹. Thus, all positions, not just those in the tails, can contribute to the functional divergence of tubulin genes.

Insights into allosteric properties of the lattice

Insight into the connection between the $\alpha\beta$ -tubulin conformational cycle and polymerization dynamics came from studies of the buried T238A mutation in the yeast β -tubulin⁵⁴. T238 mutations in β -tubulin suppress catastrophe, slow microtubule shrinking and increase spontaneous nucleation, with little effect on microtubule growth rates⁵⁴. Tubulin dimer with T238-mutated β -tubulin retains a more GTP-like conformation in the GDP lattice^{51,54}, providing a structural correlate to the observed changes in dynamics. Point mutations in β -tubulin associated with an ocular motility disorder (congenital fibrosis of the extraocular muscles (CFEOM3)) altered microtubule growth rates and/or catastrophe frequencies, even though the mutations are not found at polymerization interfaces¹³⁸, but the underlying mechanisms remain to be clarified.

a β -tubulin mutant (T238V; BOX 2), which causes microtubules to shrink more slowly, performed the same work as wild type⁵¹. This counter-intuitive result could be explained by a physical model in which the T238V mutant makes stronger lateral bonds and the shrinking rate is determined by the breaking of lateral bonds rather than by the dissociation of dimers from the microtubule tip⁵¹. But the mutated T238 residue is buried within the tubulin dimer structure and far from tubulin–tubulin interfaces, so the existence of stronger lateral bonds between neighbouring tubulins cannot be explained by direct perturbation of that interface. Instead, the stronger lateral bonds must arise from an allosteric effect in the lattice. Indeed, mutations at T238 alter the conformational cycle of $\alpha\beta$ -tubulin dimers in the lattice, favouring a more GTP-like state of the GDP-bound tubulin dimer^{51,54}. Thus, this shift in conformational preference apparently gives rise to stronger lateral bonds, but the mechanism by which it does so remains unclear. Testing hypotheses such as these will require knowledge of the structures of the mutant tubulin in the lattice. Whatever the precise structural mechanism, these optical trapping studies demonstrate that the balance of $\alpha\beta$ -tubulin conformations in the

lattice influences microtubule dynamics by tuning the strength of lateral bonds.

Together, these studies demonstrate that the $\alpha\beta$ -tubulin conformational cycle is directly linked to mechanics, on the level of both individual tubulins and the collective properties of the lattice. Microtubules push and pull, and the energetics of individual dimers are altered by the resulting compressive and tensile stresses.

MAPs impact the conformational cycle

The conformational cycle of tubulin is directly linked to microtubule dynamics. Microtubule dynamics also depend on the size and composition of the stabilizing cap, loss of which leads to microtubule catastrophe and allows dynamic instability⁶. In recent years, studies of the EB proteins and other MAPs^{4,11,12,19,55–60}, enabled by improved fluorescence microscopy assays¹⁰, have provided a wealth of new insights into the size, composition and dynamics of the stabilizing cap. These studies have revealed that several MAPs take advantage of the tubulin conformational cycle to regulate microtubule dynamics. Yet knowledge of the distributions of tubulin conformations and nucleotide states in the stabilizing cap and how they relate to each

Baculovirus

A virus that infects invertebrate cells, engineered versions of which are used to direct the expression of heterologous proteins in insect cells.

Allosteric effect

Describes a phenomenon in which the enzymatic or binding activity of one region of a protein can be influenced by binding events at other regions.

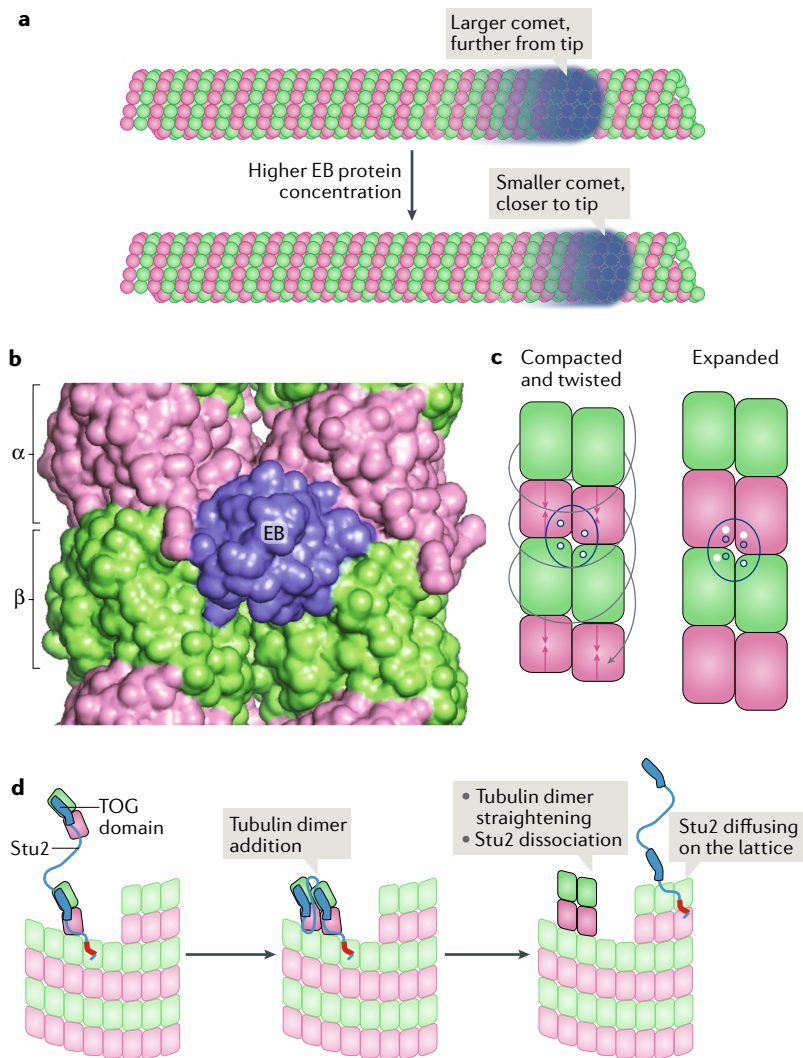


Fig. 2 | Regulatory proteins and the tubulin conformation cycle. **a** | End-binding (EB) proteins form a ‘comet’ (blue glow) near but somewhat behind the growing microtubule end (top). Increasing the concentration of EB proteins (arrow) reduces the size of the comet and moves it closer to the growing end (bottom). **b** | EB protein (blue) binds at a vertex of four $\alpha\beta$ -tubulin dimers (α : pink; β : green) in the lattice. This panel was made using PDB 3JAK. **c** | EB protein binds most tightly to a ‘compacted and twisted’ conformation of $\alpha\beta$ -tubulin, as in this conformation. The tubulin-binding epitopes on EB (round blue frames) overlap with the EB protein binding sites on tubulins (white circles). EB protein binds poorly to the expanded lattice because the EB-contacting epitopes are improperly spaced. **d** | Conformation-based mechanism of a microtubule polymerase. Multiple connected TOG domains such as those in Stu2 (only two of its TOG domains are shown) bind selectively to curved tubulin (free tubulin dimers and curved tubulins of the microtubule taper) and increase the rate of tubulin–microtubule association by concentrating unpolymerized tubulin near the microtubule end via a tethering mechanism (left). The tethered TOG-bound tubulin associates more quickly with the microtubule end, and lateral interactions between TOG-bound tubulins on the microtubule end (middle) lead to straightening, which releases the TOG domains (right) to bind curved tubulins for another round of incorporation. The red segment indicates a basic region that mediates curvature-independent lattice binding that helps retain the polymerase near the microtubule end. Part **d** is adapted from REF.⁶¹

other remains an important gap in our understanding of mechanisms governing microtubule dynamics. In this section, we discuss how MAPs interact with the mechanical cycle of tubulin, providing evidence that biochemical regulation of microtubule dynamics

through MAPs is strongly tied to the conformational changes of tubulin dimers.

Interplay between tubulin conformation and EB proteins.

Plus end tracking by an EB protein was first reconstituted in vitro using microtubule integrity protein Mal3, the EB orthologue in *Schizosaccharomyces pombe*¹⁹. Mal3 marked a ‘comet’ region hundreds of nanometres in length and also increased the frequency of microtubule catastrophe¹⁹. Similar observations followed for vertebrate EB proteins⁵⁷. Single-molecule and kinetic analyses revealed the high-affinity EB-binding site on the microtubule to be transient, with a lifetime of about 10 seconds¹⁹. Simultaneous imaging of the comet and the microtubule demonstrated that the comet region does not extend all the way to the very end of the growing microtubule¹¹. Instead, there is an ‘EB-free’ zone at the very tip¹¹ (FIG. 2a) that might overlap with the partially curved, tapered region²⁵. The presence of this zone indicates that the high-affinity EB-binding site takes time to develop and is linked to the integration of the tubulin dimers into the lattice.

Interestingly, increasing the concentration of EB decreases the size of the EB comet and causes it to shift closer to the microtubule end¹¹ (FIG. 2a), thereby reducing the size of the EB-free zone. Thus, it seems clear that EB proteins both recognize particular sites on the lattice and influence the creation and destruction of these sites. Reduction of the EB comet size is linked to increased frequency of catastrophe. These EB concentration-dependent effects have led to the hypothesis that EB proteins are maturation factors that increase the rate of GTP hydrolysis in the lattice^{11,24}. This hypothesis explains how increasing the concentration of EB can increase the catastrophe frequency because faster GTP hydrolysis should lead to smaller stabilizing caps. By revealing and quantifying a transient and distinctive region at the growing end of microtubules that is marked and altered by EB proteins, these studies provided new insight into the molecular events occurring near the growing microtubule end.

Although the dynamics of EB comets have been measured with increasing accuracy and precision¹¹, the nature of the high-affinity EB-binding site remains poorly defined. Important clues have come from experiments using microtubules containing GTP analogues. GTP γ S is a hydrolysis-resistant GTP analogue that is thought to mimic the GDP- P_i tubulin state. EB proteins bind with comparable affinity to comets of dynamic microtubules and to microtubules containing GTP γ S⁵⁸. Accordingly, GTP γ S-containing microtubules are considered to provide a model for the comet region of growing microtubules. By contrast, EB proteins bind less tightly to microtubules containing guanosine-5’-[(α,β)-methylene]triphosphate (GMPCPP), a different hydrolysis-resistant GTP analogue that stabilizes the expanded conformation of tubulin in the lattice²⁹. GMPCPP is thought to provide the best model for the GTP state of the lattice because it supports microtubule polymerization better than GTP γ S or other analogues⁶¹. These experiments have been taken to suggest that EB proteins bind preferentially to the GDP- P_i state over the GTP state.

Kinesin 13

Kinesin family proteins that act as microtubule depolymerases. These kinesins use ATP hydrolysis to induce outward curvature in protofilaments, triggering catastrophes. An example member of this family is mitotic centromere-associated kinesin (MCAK).

Kinesin 8

Family of kinesins that combines the motility of conventional kinesins with depolymerase activity. Because of this unique combination, some members of the family have been shown to depolymerize microtubules in a length-dependent manner. A well-studied member of the family is yeast kinesin-like protein Kip3.

Rescue factors

Regulatory factors that increase the likelihood of microtubule rescue. The best-studied rescue factors belong to the CLASP family.

The first cryo-EM reconstruction of EB-bound GTP γ S microtubules showed that EB proteins bind at the vertex of four $\alpha\beta$ -tubulin dimers (FIG. 2b) and that this binding is sensitive to the conformation of the tubulin dimers³⁹ (FIG. 2c). At higher resolution, $\alpha\beta$ -tubulin in the EB-bound GTP γ S lattice adopted a 'twisted and compacted' conformation that differs slightly from the compacted, GDP-bound conformation with respect to the relative rotation between $\alpha\beta$ -tubulins along a protofilament²⁴ (FIG. 1d). Consistent with this structural difference, EB proteins bind to the GDP lattice about tenfold less tightly than they bind to the comet region or GTP γ S microtubules⁵⁸. The EB–GTP γ S structure may indicate that EB proteins prefer the twisted and compacted tubulin state and bind preferentially to a post-GTPase nucleotide state of the lattice such as GDP–P_i²⁴. However, whether GTP γ S faithfully mimics a GDP–P_i state is not known. EB binding has also been observed to influence the conformation of tubulin in the lattice³⁶. Thus, EB proteins can both detect and induce the twisted and compacted tubulin state. Whether the mode of interaction of EB proteins with tubulin is conserved in all species remains unresolved: the structure of microtubules derived from *S. pombe* tubulin bound to Mal3 did not show compaction³⁷. Another caveat in defining the EB–tubulin interaction is that EB proteins have some affinity for all lattices, irrespective of their compaction state^{58,60}, a fact that may explain why EB proteins have been observed by cryo-EM to bind within the partially curved microtubule end structures²⁵ that were thought to be an EB-free zone.

The model whereby EB proteins preferentially bind to a post-GTPase state of the lattice can explain the presence of the EB-free zone at the very end of a microtubule because in this zone tubulin would be in its expanded, GTP-bound conformation, not the compacted and twisted state (FIG. 1d). However, this post-GTPase hypothesis of EB binding is at odds with the observation that increasing the concentration of EB proteins decreases the size of the EB comet¹¹; indeed, this effect on comet size has been proposed to occur because EB protein binding increases the rate of GTP hydrolysis in the microtubule^{11,24}. However, to increase the rate of GTP hydrolysis, EB proteins should bind to the lattice before hydrolysis (pre-GTPase), not after. Thus, there is an apparent conflict between the preference of EB proteins for a twisted and compacted GTP γ S lattice (supposedly a post-GTPase phenomenon) and the concentration dependence of EB comet size (rationalized as a pre-GTPase phenomenon). One way to resolve this conflict is to move away from the viewpoint that the conformation of tubulin is determined only by its nucleotide state. As discussed in the final section of the article, we argue that long-range mechanical coupling between subunits in the lattice could explain how EB proteins can stimulate GTP hydrolysis while binding preferentially to what appears to be a post-GTPase state of the lattice.

Interplay between tubulin conformation and other MAPs. EB proteins currently provide one of the best-understood regulatory mechanisms for microtubule dynamics, but recent studies of other MAPs indicate that, like EB proteins, they bind selectively to specific

conformations of tubulin and thereby modulate the tubulin conformation cycle.

Whereas EB proteins modulate the transitions between expanded and compacted conformations within the microtubule lattice, a growing number of proteins regulate transitions between straight and curved conformations of $\alpha\beta$ -tubulin at the growing microtubule end. The first protein shown to bind preferentially to the curved conformation of $\alpha\beta$ -tubulin was stathmin (STMN1)^{62,63}, which uses this activity to sequester unpolymerized $\alpha\beta$ -tubulin. It is now clear that microtubule depolymerases of the kinesin 13 family also bind preferentially to, and can induce, curved $\alpha\beta$ -tubulin⁶⁴ on the microtubule end^{65,66}, thereby triggering catastrophe^{67–69}. Binding preferentially to curved tubulin may be a universal feature of microtubule depolymerases. Indeed, a recent study of the cooperative microtubule depolymerase kinesin-like protein Kip3 (REFS^{70–72}) from the kinesin 8 family indicates that it also interacts preferentially with curved $\alpha\beta$ -tubulin⁷³ (BOX 2) and that this property underlies its depolymerase activity⁷³. Despite sharing the same overall kinesin fold, Kip3 and mitotic centromere-associated kinesin (MCAK) evolved distinct ways to bind preferentially to curved $\alpha\beta$ -tubulin^{64,73–75}: the two depolymerases use different, family-specific loop elements to bind tightest to the curved $\alpha\beta$ -tubulin. These studies emphasize that microtubule dynamics depend on the balance between straight and curved conformations on the microtubule end and demonstrate that depolymerases act by favouring curved conformations.

Surprisingly, preferential binding to curved $\alpha\beta$ -tubulin can also be used to promote microtubule growth. Indeed, microtubule polymerases in the Stu2/XMAP215 family make microtubules grow more quickly^{76–80} using TOG domains that bind preferentially to curved $\alpha\beta$ -tubulin^{81,82} (FIG. 2d). Recent advances in the mechanistic understanding of these proteins further illuminate the central role of mechanical coupling and conformational transitions in microtubule dynamics and regulation.

In the absence of unpolymerized $\alpha\beta$ -tubulin, isolated polymerase TOG domains will depolymerize stabilized microtubules, presumably by stabilizing a curved, more dissociation-prone conformation of $\alpha\beta$ -tubulin on the microtubule end⁵⁴. Yet, when part of the intact protein, multiple TOG domains catalyse polymerization in the presence of unpolymerized $\alpha\beta$ -tubulin. How can this be? While unpolymerized $\alpha\beta$ -tubulin is curved and binds tightly to TOG domains, lattice interactions between tubulins induce straighter tubulin conformations that bind weakly to TOG domains. As long as the straightening-inducing interactions in the lattice are stronger than the TOG– $\alpha\beta$ -tubulin interactions they compete with, the TOG domains will 'hand-off' their bound tubulin dimers to the lattice, allowing the polymerase to promote growth by accelerating the delivery of new subunits.

These examples drew mainly on studies of how selected MAPs control microtubule dynamics by influencing and being influenced by the conformation of tubulin subunits. It will be fascinating to discover whether related logic underlies the activity of less understood regulatory proteins such as kinesin family polymerases^{83–86}, TOG-based rescue factors (for example,

cytoplasmic linker associated proteins (CLASPs)^{87,88}, anti-catastrophe factors such as targeting protein for Xklp2 (TPX2)(see below) and others^{89,90}. Combining these MAPs together with $\alpha\beta$ -tubulin mutants that have altered conformational cycles⁵⁴ (BOX 2) should yield more

direct insight into the mechanical underpinnings of the regulation of microtubule dynamics.

Implications for nucleation and rescue

In the previous sections, we focused on how the conformational cycle of tubulin dimers is linked to microtubule growth and catastrophe. Notably, similar processes also operate during microtubule nucleation^{91,92} and rescue⁹³.

Microtubules can nucleate spontaneously (form de novo) when unpolymerized $\alpha\beta$ -tubulin oligomerizes in solution (FIG. 3a). Oligomers below a certain size are unstable, but growth becomes favourable after a critical nucleus is formed^{94,95}. Because tubulin oligomers grow from soluble GTP-tubulin, which is curved, these small oligomers must also have curvature and may resemble growing microtubule ends. The kinetics of microtubule nucleation are very sensitive to the concentration of unpolymerized tubulin^{96,97}, suggesting that the critical nucleus requires multiple, kinetically unfavourable steps to form.

Cells are thought to bypass these unfavourable steps by using templates such as the γ -tubulin ring complex (γ -TuRC)^{98,99} or newly severed microtubule ends (for example, REF.¹⁰⁰) to nucleate microtubules. In vitro, templates such as the γ -TuRC or stabilized microtubule 'seeds' accelerate microtubule formation and reduce the tubulin concentration required to observe elongation, but this templated nucleation still shows a time lag¹⁰¹. The presence of this lag, which is unexpected¹⁰², indicates that templates imperfectly mimic the microtubule end, which is the preferred substrate for microtubule polymerization (FIG. 3b).

Why might blunt templates be imperfect substrates for elongation compared with a growing microtubule end? One source of imperfection may be that templates such as the γ -TuRCs are blunt rings rather than the tapered, variably curved arrangements of $\alpha\beta$ -tubulin normally found at growing microtubule ends (FIG. 3b; BOX 1). Tubulin dimers may associate onto a blunt template with lower affinity because blunt templates require more straightening of the incoming subunits and/or because blunt templates lack the 'raggedness' that generates favourable corner binding sites. The maturation of a blunt template into a tapered, variably curved microtubule end (BOX 1) can be accelerated in vitro by MAPs that interact preferentially with partially curved tubulin conformations that stabilize $\alpha\beta$ -tubulin- $\alpha\beta$ -tubulin interfaces or that accelerate the association of $\alpha\beta$ -tubulin¹⁰¹.

Several MAPs have been shown to regulate nucleation (for example, REFS^{103–105}), and new data show that they take advantage of the tubulin conformation cycle to do so. XMAP215 family proteins provide an example of how accelerating the binding of $\alpha\beta$ -tubulin to microtubule ends can promote nucleation^{101,105,106}, which depends on conformation-specific interactions of XMAP215 with tubulin dimers (FIG. 3c, left). More specifically, XMAP215 family proteins may tether curved tubulin dimers to early nucleation intermediates. TPX2 is a spindle assembly factor in the chromatins-mediated microtubule nucleation pathway^{107–109}. TPX2 suppresses catastrophe and slows post-catastrophe shrinking in vitro^{101,106,110}. TPX2 also reduces the lag for templated nucleation¹⁰¹, which may be critical to its function given

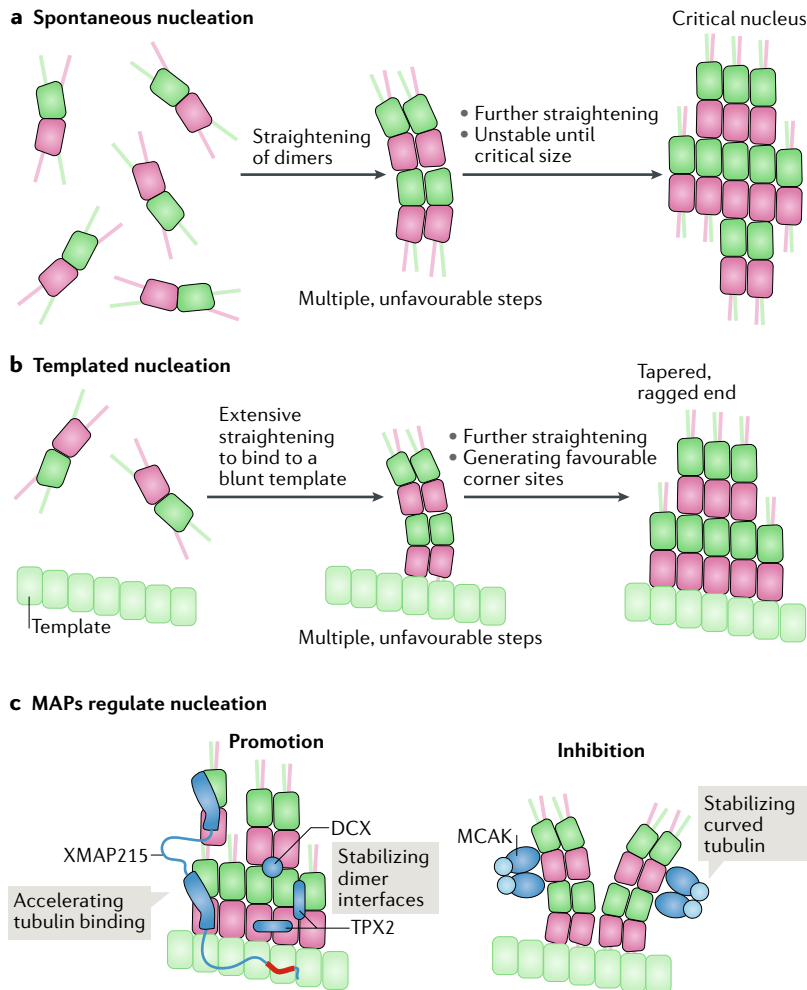


Fig. 3 | Barriers to spontaneous and templated nucleation. a | Spontaneous nucleation involves multiple, unfavourable steps. Unpolymerized tubulin is curved (left) and must straighten to form small oligomers (middle). The growth of these oligomers requires increased straightening of newly added tubulins and of those already in the oligomer. These early interactions are unstable; thus, high concentrations are needed to progress through these unfavourable intermediates to form a critical nucleus that supports further expansion of the lattice (right). **b** | The presence of a template for nucleation stabilizes initial unfavourable interactions between tubulin dimers. Nevertheless, templated nucleation still involves multiple, unfavourable steps, owing to the bluntness of the template (left). Curved tubulin binding to the blunt template may have to straighten more than at the partially curved microtubule end (middle), and the transition of the blunt template into a growing microtubule end requires many tubulins to straighten (right). A blunt template also presents fewer high-affinity corner sites than a tapered growing end (BOX 1; FIG. 1b), thereby making addition of new tubulin dimers less favourable than at the growing microtubule end. **c** | Microtubule-associated proteins (MAPs) regulate microtubule nucleation by recognizing and altering the conformation of tubulin. Nucleation-promoting MAPs such as XMAP215, targeting protein for Xklp2 (TPX2) and neuronal migration protein doublecortin (DCX) help form a nascent plus end (left). They recognize curved tubulin conformations present at the microtubule tip and accelerate addition of $\alpha\beta$ -tubulin (XMAP215; only two of its TOG domains are shown) or stabilize tubulin-tubulin interaction at interfaces between dimers (TPX2 and DCX). Nucleation-inhibiting MAPs such as mitotic centromere-associated kinesin (MCAK) destabilize the microtubule end by binding and promoting highly curved tubulin conformation (right).

TPX2

A spindle assembly factor that functions in the Ran–GTP pathway. TPX2 is released from importins in the vicinity of chromosomes, where it stimulates microtubule nucleation and interacts with motor proteins.

Microtubule nucleation

The process of forming a microtubule *de novo* (from unpolymerized $\alpha\beta$ -tubulin subunits) or on a template such as the γ -tubulin ring complex or a severed microtubule end.

γ -tubulin ring complex

(γ -TuRC). A conical oligomer of γ -tubulins and γ -ring complex proteins (GCPs) that creates a template for microtubule nucleation.

Chromatin-mediated microtubule nucleation pathway

A pathway for microtubule nucleation wherein factors that promote nucleation are selectively activated in the vicinity of chromatin.

Branching nucleation

A microtubule-dependent microtubule nucleation pathway wherein new microtubules are nucleated from the sides of existing ones.

that TPX2 also interacts directly with γ -TuRCs during branching nucleation^{111,112}. TPX2 can also stabilize stubs (small oligomers) of tubulin that act as nuclei for microtubule formation *in vitro*¹⁰⁶. Suppressing catastrophe and stabilizing small oligomers of $\alpha\beta$ -tubulin indicates that TPX2 stabilizes an $\alpha\beta$ -tubulin– $\alpha\beta$ -tubulin interface (FIG. 3c, left), and a recent cryo-EM structure showed TPX2 binding across both longitudinal and lateral interfaces¹¹³. Interestingly, TPX2 also binds preferentially to partially curved tubulin structures at microtubule ends¹⁰⁶. The neuronal migration protein doublecortin (DCX), which is expressed in developing neurons and is unrelated to TPX2, provides another example of a MAP that regulates nucleation *in vitro*¹¹⁴. DCX also binds to partially curved tubulin structures¹¹⁵ by binding at the vertex of four $\alpha\beta$ -tubulins¹¹⁶ (FIG. 3c, left). DCX nucleates microtubules spontaneously with extraordinary effectiveness^{117–119}. Thus, two MAPs that stabilize $\alpha\beta$ -tubulin– $\alpha\beta$ -tubulin interfaces in partially curved tubulin assemblies at microtubule ends show strong nucleation activity, indicating that microtubule ends and nucleation intermediates are structurally similar. Interactions of MAPs with conformations found in these intermediates may thus be the key to regulating nucleation.

Depending on the conformation being stabilized, MAPs can also inhibit nucleation. For example, end-binding protein 1 (EB1; also known as MAPRE1) and MCAK each reduce the rate of templated nucleation and increase the concentration of tubulin required to observe elongation¹⁰¹. As discussed above, EB1 may accomplish this by promoting a twisted and compacted state of $\alpha\beta$ -tubulin²⁴ in the GTP lattice that may have higher GTPase activity and/or be a less favourable substrate for elongation. MCAK antagonizes nucleation because it prefers fully curved conformations of tubulin that are not compatible with the variably curved structures present at the growing microtubule end and in nucleation intermediates (FIG. 3c, right).

The lag associated with nucleation on a blunt template thus reflects an energetic barrier to develop a properly structured microtubule end. By modulating what must be a delicate balance between different conformations of tubulin in the lattice, MAPs can either promote or inhibit this development.

The transition from a blunt template to a true, persistently growing microtubule end faces a barrier. There must be an analogous, larger barrier to rescue when the outwardly peeling protofilaments convert back into a growing microtubule end. This barrier to rescue may explain why rescues are rarely observed *in vitro*. Rescues are readily observed in cells, however, likely owing to the presence of rescue factors. Perhaps the best-characterized rescue factor is Cls1 (also known as Peg1), the CLASP family protein in *S. pombe*⁸⁸. CLASP family proteins⁸⁷ contain TOG domains, but some of these TOG domains seem likely to prefer a different conformation of tubulin¹²⁰ compared with the microtubule polymerase TOG domains that preferentially bind to curved dimers; thus, CLASPs may facilitate rescue by interacting with a specific conformation of tubulin that occurs during protofilament peeling. However, little is currently known about how

the CLASP TOG domains actually contribute to rescue or about how peeling protofilaments convert back into straighter ones.

An unexpected connection between rescue and microtubule mechanics comes from recent work examining how microtubules respond to bending stress¹²¹, which again exemplifies how mechanical forces can alter the biochemical properties of tubulin in the lattice. In this work, microtubules were bent repeatedly by controlled fluid flow in a microfluidics chamber; the microtubule became increasingly flexible with each bend. After repeated bending, the baseline stiffness of the microtubule could be restored over time in the presence of soluble tubulin; this recovery in stiffness suggested that the mechanical stress from repeated bending had damaged the lattice and that fresh tubulin could enter and repair these damaged sites. Sites of photodamage can also be repaired by fresh tubulin¹²¹. Interestingly, microtubule rescues cluster at sites of repair *in vitro*¹²². In cells, sites of damage also appear to undergo repair and facilitate rescue¹²². Cells may also protect their microtubules from damage by acetylating tubulin^{123,124}, but whether and how acetylation is linked to modulating microtubule mechanics remains unclear. Although we do not understand what repair entails, rescues may occur near repaired sites because the fresh tubulin stabilizes the region and reduces or eliminates protofilament peeling, thereby providing a favourable substrate for polymerization.

Under normal circumstances, regardless of nucleotide state, tubulin below the cap and in the body of the lattice is thought to be so tightly integrated into the lattice by tubulin–tubulin interactions that it does not dissociate appreciably. By demonstrating that the mechanical deformations accompanying bending can considerably alter how strongly tubulin is bound in the lattice and lead to local tubulin dissociation, this line of work provides clear new evidence for the importance of mechanics in regulating microtubule structure and dynamics.

Mechanical coupling in the lattice

The complex structure of the microtubule end and the presence of expanded, compacted and variably curved conformations of tubulin therein that differently interact with each other and that are modulated by MAPs raise a number of questions about how the conformational states of tubulin affect the biochemistry of tubulin–tubulin interfaces and vice versa. For example, what drives the straightening of curved GTP-tubulin dimers when they assemble into the lattice at the growing end or in nucleation intermediates? It has been proposed that straightening would be driven by the alignment of lateral interfaces between neighbouring dimers³⁴. The presence of variably curved conformations at the end implies that the straightening of one dimer both requires and facilitates the straightening of its neighbours¹²⁵ (FIG. 4a). These interactions describe a kind of allostery, wherein the effect of one tubulin binding to another is transmitted to other interfaces via changes in curvature.

Likewise, how does the compaction of a single tubulin dimer affect its neighbours? The expanded GTP-tubulin lattice and compacted GDP-tubulin lattice

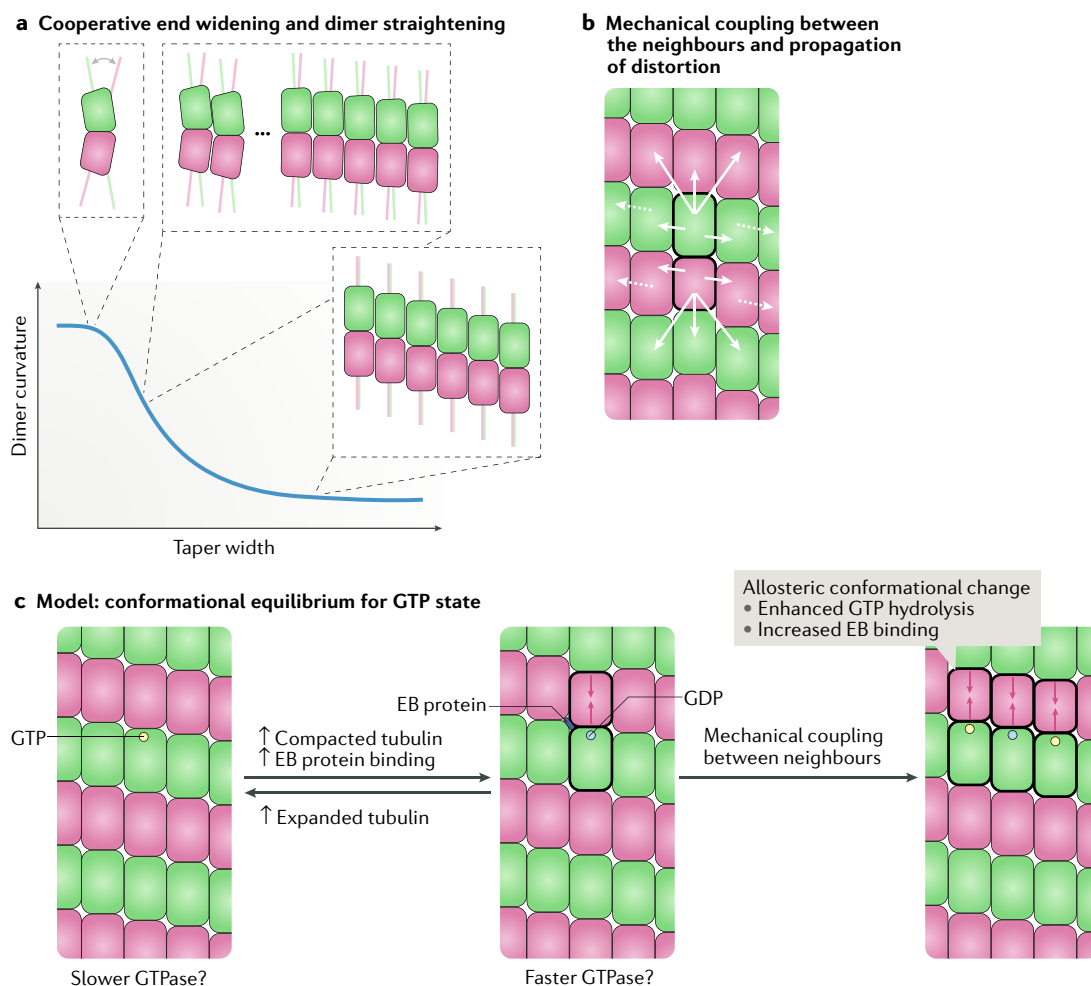


Fig. 4 | Mechanical coupling in the microtubule lattice. a | At the microtubule end or in nucleation intermediates, tubulin curvature is thought to vary with the width of the taper or oligomer — in wider tapers or oligomers, the tubulin dimers are straighter and the strength of tubulin–tubulin interactions is altered. Coloured lines on the $\alpha\beta$ -tubulin cartoons provide a visual reference for the degree of curvature. Curvature change must propagate through the lattice, suggesting that individual dimers are coupled and that this coupling can support correlated rearrangements of multiple tubulins. **b** | Cartoon illustrating a rationale for mechanical coupling in the lattice. A compacted tubulin (middle) is shown in a majority expanded lattice. The mismatch between expanded and compacted conformations is likely resolved through conformational changes near the mismatch (white arrows). The periodic nature of the lattice means that such accommodation could propagate beyond nearest neighbours (dashed arrows). **c** | Model for conformational control of GTPase activity of tubulin that was inspired by observations of end-binding (EB protein)–microtubule interactions. The model speculates that because EB proteins bind less efficiently to expanded tubulins than to compacted tubulins (FIG. 2c), the GTPase activity is slower in the expanded versus the compacted state. A compacted state in an otherwise expanded lattice will distort neighbouring tubulins owing to long-range coupling between tubulins (see also part **b**), which may enhance their GTPase activity and thereby promote further EB binding by favouring compaction. Such a model is expected to display switch-like, concerted behaviours of multiple tubulins and could give rise to cooperative, positive-feedback enhancement of GTPase activity in the stabilizing cap, possibly explaining EB protein–tubulin interaction dynamics derived experimentally.

are incommensurate²⁹. The mismatch between these conformations must introduce structural conflicts near the microtubule end where the lattice contains a mixture of nucleotide states. Resolution of these conflicts, wherein tubulin dimers change conformation to accommodate their mismatched neighbours, may give rise to long-range coupling in the lattice: the effects of a conformational transition in one tubulin may propagate beyond its immediate neighbours to alter the conformation and/or biochemistry of more distant subunits in the lattice. For example, when one $\alpha\beta$ -tubulin compacts in an otherwise expanded GTP lattice, the resulting structural mismatch

likely creates a kind of stress that makes neighbouring tubulins more likely to compact, which would in turn make their neighbours more likely to compact, and so on (FIG. 4b). This conformational coupling could in principle override or modulate the effect of nucleotide state on tubulin conformation. Because of this mechanical coupling in the lattice, the conformation of an individual $\alpha\beta$ -tubulin might not be uniquely determined by its nucleotide state but rather also by the conformation of its neighbours. Such long-range mechanical coupling would make the microtubule end a highly cooperative structure capable of abrupt transitions.

Long-range coupling in the lattice provides a new way to think about how regulatory proteins such as EB proteins could interact with microtubules and influence tubulin conformation in the lattice. Recall the pre-GTPase versus post-GTPase paradox: EB proteins appear to accelerate the rate of GTP hydrolysis while binding preferentially to what has been thought to be a post-GTPase conformation in the lattice. These behaviours are hard to reconcile with a view in which conformation of an individual $\alpha\beta$ -tubulin is uniquely determined by its nucleotide state. Long-range coupling in the lattice may resolve this paradox (FIG. 4c). Some evidence for the existence of long-range coupling comes from an interesting recent study of XMAP215 and EB1 that revealed synergistic action of the two MAPs: even though EB1 and XMAP215 did not interact directly, the presence of EB1 made XMAP215 a more potent polymerase. This synergy was proposed to result from long-range coupling in the lattice¹²⁶. Indeed, when EB proteins bind to and stabilize a twisted and compacted conformation at one site, long-range coupling means the EB-induced change in conformation could propagate some distance through the lattice. If the corresponding changes in conformation also elevate the rate of GTP hydrolysis at remote sites in the stabilizing cap, then preference of EB proteins for a twisted and compacted conformation at one site has the potential to accelerate GTP hydrolysis elsewhere. This hypothesis should ultimately be testable using $\alpha\beta$ -tubulin mutants with altered conformational cycles. The idea that long-range mechanical coupling might influence GTP hydrolysis throughout the stabilizing cap also has implications for the mechanism of microtubule catastrophe. Mechanical coupling creates the possibility for positive feedback and switch-like effects, wherein the compaction state and/or GTPase rate could change abruptly across a number of coupled $\alpha\beta$ -tubulin dimers. For example, consider that the GTPase rate may be activated by compaction, but compaction of any one tubulin is disfavoured if most of its neighbours are expanded. In this scenario, a threshold amount of GTP hydrolysis may be required before appreciable compaction can occur; upon reaching this threshold, the cap would be

abruptly lost (like a balloon popping) because a zone of tubulins would compact at the same time. Other models that do not invoke compaction-activated GTPase activity are also possible. Whatever the mechanism, we speculate that long-range coupling in the lattice and associated threshold-dependent effects will be important for understanding fundamental mechanisms of microtubule dynamics.

Conclusion and perspective

The microtubule field has made remarkable progress advancing the biochemical and mechanical understanding of microtubule polymerization dynamics. New structural states of $\alpha\beta$ -tubulin have been defined in atomic detail^{24,29}, and we are beginning to appreciate how the transitions between these states create the phenomenon of dynamic instability⁵⁴ and how MAPs can manipulate these transitions to modulate microtubule dynamics^{73,74,81,82,90,113}. Studies of the material properties of microtubules — how they generate and respond to force and transmit information over nearest-neighbour distance and longer distances — have also greatly increased our understanding of microtubule dynamics^{42,51,126}. However, we still lack an understanding of how biochemistry determines mechanics and of how mechanics modulates biochemistry. Integrating these two perspectives remains a major challenge for the future.

An integrated understanding of microtubule dynamics and regulation that unifies mechanical and biochemical perspectives will require deeper insight into the conformations of $\alpha\beta$ -tubulin and their relative energetics, new knowledge of how regulatory factor binding can modulate the distribution of $\alpha\beta$ -tubulin conformations and the ability to detect and quantify the contribution of mechanical coupling between tubulin dimers in the lattice. Although formidable challenges remain, the remarkable recent leaps forward leave us optimistic about the prospects for substantial progress in understanding microtubule dynamics on a molecular scale in the near future.

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